Zewail City of Science, Technology and Innovation
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Brazilian E-Commerce Project Report

[Phase One : Choose Dataset and start working on cleaning it]

Problem Statement:

The Brazilian E-Commerce Public Dataset by Olist presents several data governance challenges that must be resolved before the data can be used for reliable analysis. These include over 157,000 missing cells (6.9%) concentrated in review comment and delivery date columns, severe class imbalance in order_status (97.1% delivered) and payment_type (73.9% credit_card), invalid payment type entries (‘not_defined’), extreme outliers in payment_value and delivery_time, date columns stored as plain strings, and data spread across 8 separate relational files requiring careful merging.

This project documents the full data governance pipeline applied to address each of these issues.

Data Description:

The Brazilian E-Commerce Public Dataset by Olist is a real-world commercial dataset containing over 100,000 orders placed on the Olist Store between September 2016 and October 2018, capturing the full customer journey from purchase to delivery and review. For this project, four relational tables (orders, customers, payments, and reviews) were merged into a single analytical dataset of 103,677 records and 22 columns, covering order timelines, customer demographics (city, state, ZIP code), payment details (type, value, installments), and review feedback. The dataset is publicly available on Kaggle and exhibits several data quality issues that required resolution before analysis.

Data Profiling:

The team applied automated statistical profiling using the ydata-profiling library (v4.18.1) in Python. This tool generates a comprehensive HTML report covering variable types, distributions, missing values, correlations, duplicates, and data quality alerts. The full profiling report was generated on April 11, 2026 and took 19.23 seconds to complete.

The profiling was performed on the merged dataset (all 4 tables joined) rather than individual tables, to capture the full picture of data quality issues in context.

Alert	Type	Description
review_comment_title	Missing	91,681 missing values — 88.4% of the column is empty
review_comment_message	Missing	60,862 missing values — 58.7% of customers leave no comment
order_delivered_customer_date	Missing	3,030 missing — orders not yet delivered at data collection time
order_delivered_carrier_date	Missing	1,861 missing — orders not picked up by carrier yet
order_status	Imbalance	91.5% imbalance — 97.1% of all records have status 'delivered'
payment_type	Imbalance	52.6% imbalance — credit_card dominates at 73.9%
customer_state	High Correlation	Highly correlated with customer_zip_code_prefix (geographic dependency)
customer_zip_c	High Correlation	Highly correlated with customer_state (expected geographic link)ode_prefix

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Data Cleaning:

The cleaning pipeline was implemented in Python using pandas and followed seven steps. First, the dataset was inspected using df.info() and df.isnull().sum() to establish a baseline. Four columns were then dropped: order_id and customer_id (surrogate keys not needed after merging), and review_comment_title and review_comment_message (88.4% and 58.7% missing respectively). Three records with payment_type = ‘not_defined’ were removed as data entry errors, and 171 null values in order_approved_at were filled using ffill().bfill(). All five date columns were converted from strings to pandas datetime objects. A new delivery_time feature was derived as the difference in days between purchase and delivery. Finally, IQR-based outlier removal was applied to delivery_time for delivered orders only, ensuring non-delivered orders with null dates were not incorrectly filtered out.

Results:

After applying all cleaning steps, the dataset was reduced from 103,677 to 98,527 rows and from 22 to 19 columns (4 dropped, 1 feature added). Total missing cells fell from 157,605 to 4,875, no duplicate rows were found, the 3 invalid payment_type records were removed, and the 171 null approval dates were filled. The remaining 4,875 nulls are intentional and reflect real-world business logic: 1,856 missing values in order_delivered_carrier_date and 3,019 each in order_delivered_customer_date and delivery_time correspond to orders that had not yet been delivered at the time of data collection and are therefore expected to have no delivery dates.

Data Validation and Schema:

A validation schema was developed to ensure the cleaned dataset meets required data quality standards. The schema defines rules for each column covering data type, nullability, value ranges, allowed categories, and logical consistency constraints.

Column	**Type**	**Nullable**	**Validation Rules**
order_status	Categorical	No	Allowed values: {delivered, shipped, canceled, unavailable, invoiced, processing, created, approved}
order_purchase_timestamp	DateTime	No	Must be a valid datetime Range: 2016-09-01 to 2018-12-31 Must be <= order_approved_at
order_approved_at	DateTime	No	Must be >= order_purchase_timestamp No future dates beyond dataset range
order_delivered_carrier_date	DateTime	Yes	If not null: must be >= order_approved_at Only null if order_status != delivered
order_delivered_customer_date	DateTime	Yes	If not null: must be >= order_delivered_carrier_date Only null if order_status != delivered
order_estimated_delivery_date	DateTime	No	Must be a valid datetime Must be >= order_purchase_timestamp
customer_unique_id	Text	No	Exactly 32 alphanumeric characters Matches pattern: [a-f0-9]{32}
customer_zip_code_prefix	Integer	No	Range: 1000 to 99999 No negatives or zeros
customer_city	Text	No	Length: 2 to 50 characters Lowercase string, no digits

customer_state	Categorical	No	Must be one of the 27 Brazilian state abbreviations e.g., SP, RJ, MG, RS, PR, SC, BA, DF, GO, ES, ...
payment_sequential	Integer	No	Range: 1 to 29 Positive integers only
payment_type	Categorical	No	Allowed values: {credit_card, boleto, voucher, debit_card} 'not_defined' is NOT allowed
payment_installments	Integer	No	Range: 0 to 24 0 only allowed for voucher payments
payment_value	Float	No	Must be >= 0 0 only allowed for voucher-type orders Max reasonable cap: 20,000 BRL
review_id	Text	No	Exactly 32 alphanumeric characters Matches pattern: [a-f0-9]{32}
review_score	Integer	No	Allowed values: {1, 2, 3, 4, 5} No other values permitted
review_creation_date	DateTime	No	Must be a valid datetime Should be >= order_delivered_customer_date (if not null)
review_answer_timestamp	DateTime	No	Must be >= review_creation_date
delivery_time	Float	Yes	If not null: must be > 0 Null only for non-delivered orders After IQR cleaning: no extreme outliers